

The M/M/1 Queue



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Keywords: Queuing, Markov chains, continuous time

14th August, 2024

Abstract

Simulations of an M/M/1 queue with finite capacity are performed. The probability that the queue has different lengths is studied. A comparison between the distribution for the queue size obtained by sampling and the distribution that can be derived using the theory of continuous-time Markov chains is performed. Next, a study is conducted to investigate how the total time spent queuing and being served depends on the rate of arrivals. We once again find that the result obtained from simulations matches the analytic result for the expectation that can be derived.

1 Introduction

Queues appear whenever a company has a limited set of resources for providing a service to customers. Customers generally do not like to wait in long lines. Models that allow companies to understand how the lengths of queues for their services depend on the factors they can control are thus enormously beneficial. Companies can use ideas from the mathematical discipline known as queuing theory to understand how a change to their business might affect the lengths of the queues customers have to wait in before they try the change within their industry.

In this report, I will study the simplest model of a queue - the so-called M/M/1 queue. I will start by explaining the assumptions that this model makes in section 2. I will then calculate the fraction of time that the queue has various lengths according to this model in section 3. I will then finish by discussing in section 4 how the amount of time a customer has to wait in the queue depends on the ratio of the average time between arrivals to the reciprocal of the average length of service.

2 The model

The M/M/1 queue is an example of a Markov chain in continuous time. In this report I will study a queue with finite capacity so there are a finite number of states in this chain. The states describe the number of people in the queue. The model represents a situation where one person is serving the people in the queue. The first state in the chain is thus state 0 and describes the case when no one is obtaining service or queuing. The second state in the chain is state one and describes a situation where one person is receiving service. States with $N > 1$

then describe a situation where one person is receiving service, and $N - 1$ people waiting to be served.

The jump rate matrix, $\mathbf{Q}$, for the M/M/1 queue is given by:

$$\mathbf{Q} = \begin{pmatrix} -\lambda & \lambda & 0 & 0 & 0 & \dots & 0 & 0 \\ \mu & -\mu - \lambda & \lambda & 0 & 0 & \dots & 0 & 0 \\ 0 & \mu & -\mu - \lambda & \lambda & 0 & \dots & 0 & 0 \\ 0 & 0 & \mu & -\mu - \lambda & \lambda & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & 0 & \dots & \mu & -\mu \end{pmatrix} \quad (1)$$

This $C + 1$ by $C + 1$ matrix is constructed by assuming that the times customers arrive can be modelled using a Poisson process with rate λ . The amount of time it takes any given customer to be served is given by an exponential random variable with parameter μ and the maximum number of people who can be in the queue is C . The model has three parameters (λ , μ and C). C is unit less and we set it to 10 in what follows. We then set $\mu = 1$ throughout this report and give values of λ in units of μ .

3 The distribution of lengths

We first set out to determine the probability distribution for the random variable that measures the number of people in the queue for the model that was introduced in section 2. λ was set equal to 0.5μ in these simulations. Figure 1 shows the distribution for the number of people in the queue that was obtained by sampling.

To generate the points that are shown as error bars in figure 1 we modelled 20 groups of 10,000 customers being served. We calculated 20 separate histograms of the fraction of time that the queue had each size for each group of individuals. The points in the figure are thus averages over these 20 separate simulations. The error bars were then calculated using the central limit theorem and indicate the 90% confidence limit on this estimate of the average.

As $\lambda < \mu$, it is possible to find a stationary distribution, π , for the M/M/1 queue that was modelled in this section. The stationary distribution for a Markov chain in continuous time is found by finding the non-trivial vector π that satisfies:

$$\pi \mathbf{Q} = \mathbf{0}$$

where $\mathbf{0}$ is a vector containing all zeros. For the M/M/1 queue, you can use the well-known formula for the sum of a geometric progression and the jump rate matrix in equation 1 to

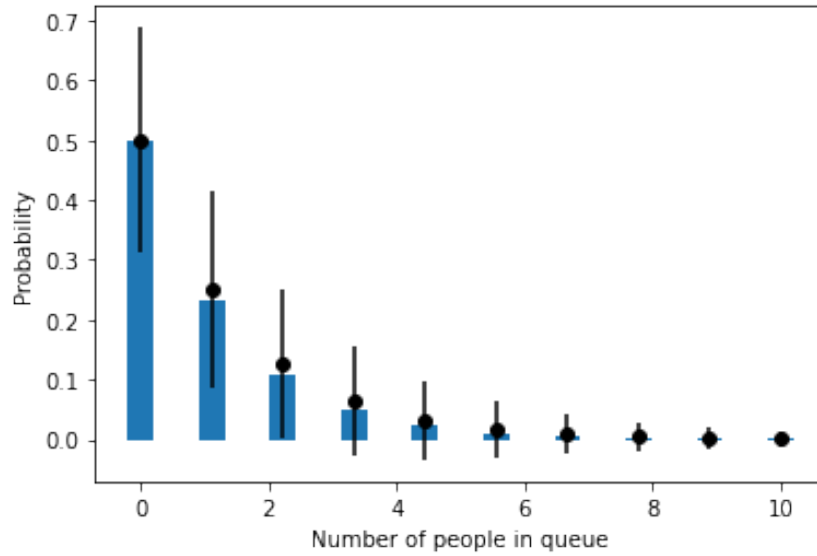


Fig. 1. The fraction of time that an M/M/1 queue with $\lambda = 0.5\mu$ has different numbers of people in it. The black points are the estimates for this distribution that were obtained by sampling. To calculate these estimates, I generated 20 histograms based on a simulation of 10000 people passing through the queue. The black points are plotted at the average from these 20 estimates, and the error bar shows a 90% confidence limit on these estimates. The blue bars indicate the probabilities of being in each state calculated using the expression for the stationary distribution of the model.

show that the elements of π are given by:

$$\pi_k = \frac{\left(1 - \frac{\lambda}{\mu}\right) \left(\frac{\lambda}{\mu}\right)^k}{1 - \left(\frac{\lambda}{\mu}\right)^{N+1}}$$

The values for the probabilities of being in the various states are shown indicated using the blue bars in figure 1. You can see that these actual values are well within the confidence limits on the estimates obtained by sampling.

4 The average time

Calculating the fraction of time that the queue has each of the possible lengths is interesting. It is probably more helpful to calculate the average amount of time each customer spends in the queue. Figure 2 thus shows how the average time it takes from the time the person enters the queue to the end of their service depends on the arrival rate λ .

The black dots in figure 2 show the averages for the total time spent queuing and being served that were obtained by sampling. The times that 100000 customers spent queuing were calculated for each value of λ that is shown in the graph. Error bars that show the 90% confidence limit on this estimate of the average were also calculated. These error bars are smaller than the sizes of the dots in figure 2.

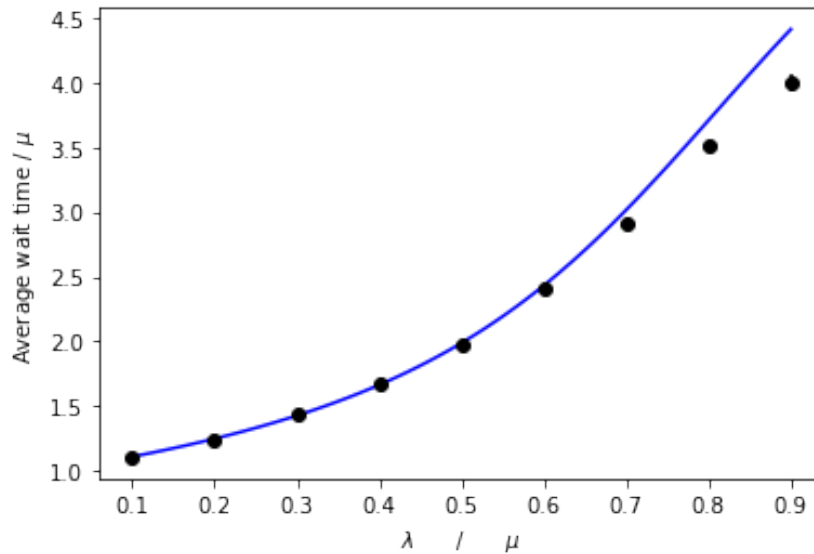


Fig. 2. The average total amount of time that customers spend queuing and being served as a function of the arrival rate. The black dots show values that were obtained by simulating a sample of 100000 customers. The 90% confidence limits on these estimates were computed, but they are smaller than the size of the black dots. The blue curve shows the value of the analytic expression for this expectation.

Little's Law tells us that the average waiting time $\mathbb{E}(W)$ that was calculated in figure 2 is related to the average number of customers in the queue $\mathbb{E}(L)$ by:

$$\mathbb{E}(W) = \frac{\mathbb{E}(L)}{\lambda}$$

$\mathbb{E}(L)$ can be calculated from the stationary distribution that was introduced in section 3 as follows:

$$\mathbb{E}(L) = \mathbb{E}(L) = \sum_{n=0}^C n \pi_n = \frac{1 - \frac{\lambda}{\mu}}{1 - \left(\frac{\lambda}{\mu}\right)^{C+1}} \sum_{n=0}^C n \left(\frac{\lambda}{\mu}\right)^n = \left(\frac{\lambda}{\mu}\right) \frac{\left(1 - (C+1) \left(\frac{\lambda}{\mu}\right)^C + C \left(\frac{\lambda}{\mu}\right)^{C+1}\right)}{\left[1 - \left(\frac{\lambda}{\mu}\right)^{C+1}\right] \left(1 - \frac{\lambda}{\mu}\right)}$$

Inserting this result into Little's Law gives:

$$\mathbb{E}(W) = \mathbb{E}(W) = \frac{\left(1 - (C+1) \left(\frac{\lambda}{\mu}\right)^C + C \left(\frac{\lambda}{\mu}\right)^{C+1}\right)}{\mu \left[1 - \left(\frac{\lambda}{\mu}\right)^{C+1}\right] \left(1 - \frac{\lambda}{\mu}\right)}$$

This analytic result for the expected waiting time is indicated using the blue curve in figure 2. As you can see, the results obtained by sampling there is a good match between the analytic result and our numerical results when λ is small. However, the numerical estimate for the

average wait time is slightly lower than the analytic value for this quantity when λ is large. The reason for this discrepancy should be investigated during further work.